14. Write a code using RAW sockets to implement packet sniffing

Code

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst

        # Determine the protocol
        protocol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = TCP
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown Protocol"
        # Print Packet details
        print(f"Protocol : {protocol_name}")
        print(f"Source IP: {src_ip}")
        print(f"Destination IP: {dst_ip}")
        print("-" * 50)
# Capture packets on the default network interface
sniff(iface = "Wi-Fi", prn = packet_callback, filter = "ip", store = 0)
```

```
------------------------------------------------------
Protocol: TCP
Source IP: 4.150.223.98
Destination IP: 172.16.11.174
------------------------------------------------------
Protocol: UDP
Source IP: 172.16.11.137
Destination IP: 224.0.0.251
------------------------------------------------------
Protocol: Unknown
Source IP: 172.16.11.194
Destination IP: 224.0.0.2
```

